

1. Project Title

Design & Build a RAG-Based Customer Support Assistant (with LangGraph & HITL)

2. Objective

The objective of this project is to develop an intelligent customer support assistant using Retrieval-Augmented Generation (RAG). The system retrieves relevant information from a document and generates context-aware responses using a language model. It also integrates Human-in-the-Loop (HITL) to handle uncertain queries.

3. System Requirements

Hardware Requirements

- Minimum 8 GB RAM
- 10 GB free storage
- Internet connection

Software Requirements

- Python 3.10+
- VS Code / Jupyter / Colab
- Required Libraries:
 - PyPDF
 - Sentence Transformers
 - NumPy
 - Transformers
 - LangGraph

4. Methodology

The system follows a step-by-step pipeline:

1. **Document Ingestion**
 - a. Load PDF and extract text
2. **Text Processing**
 - a. Split into chunks
3. **Embedding Generation**
 - a. Convert text into vectors
4. **Query Processing**
 - a. Convert user query into embedding
5. **Retrieval**
 - a. Find most relevant chunks
6. **Response Generation**
 - a. Generate answer using LLM
7. **Decision Making**
 - a. Check confidence score
8. **HITL Integration**
 - a. Escalate to human if needed

5. Implementation Details

Step 1: PDF Loading

- Used PyPDF to extract text from document

Step 2: Chunking

- Divided text into fixed-size chunks (500 characters)

Step 3: Embeddings

- Used SentenceTransformer model
- Generated embeddings for chunks and queries

Step 4: Retrieval

- Used cosine similarity (dot product)
- Selected top 3 relevant chunks

Step 5: LLM Integration

- Used HuggingFace Transformers
- Generated response using context

Step 6: LangGraph Workflow

- Defined nodes and edges
- Controlled execution flow

Step 7: HITL Implementation

- Triggered on low similarity
- Human input used as fallback

6. Features

- Context-aware question answering
- Efficient document retrieval
- Graph-based workflow (LangGraph)
- Human-in-the-loop support
- Lightweight and fast execution

7. Advantages

- Reduces manual customer support workload
- Improves response accuracy
- Scalable architecture
- Handles unknown queries via escalation

8. Limitations

- Works on single document
- Limited accuracy with small LLM
- Requires proper PDF format

9. Future Enhancements

- Integrate ChromaDB for scalable storage
- Use advanced LLM (GPT-4 / LLaMA)
- Add web interface (React + FastAPI)
- Multi-document support
- Voice-based interaction

10. Conclusion

The project successfully demonstrates the implementation of a RAG-based system for customer support. By combining retrieval techniques with language models and incorporating human feedback, the system provides accurate and reliable responses. It serves as a scalable foundation for real-world AI support systems.